

SEMESTER – I – Detailed Syllabus**Course Code: BCS101B Digital System Design****Course Learning Objectives:**

The objective of this course is:

- To introduce the fundamental concepts and methods for design of digital circuits and pre-requisite for computer organization and architecture, microprocessor systems.
- To provide the concept of designing combinational, sequential circuits and exploring the microarchitecture of computer.

Course Learning Outcomes:

After completion of the course, the learners will be able:

- To understand the basic processor unit, instruction sets and control unit.
- To analyze and minimize Boolean expressions.
- To design and analyze combinational and sequential circuits.
- To explore the memory unit and Input /Output unit.

Unit	Content	No. of Lectures:45
1	1.1 Computer Abstractions and Technology: Comparison of Computer Organization & Architecture, Computer Components and Functions, Interconnection Structures. Bus Interconnections. 1.2 Number System: Numbering system, Binary number system, Octal number system, Hexadecimal number system, Conversion from one number system to another. 1.3 Binary Adders/Subtractors: Half adder, full adder, ripple carry adder, Half subtractor, Full Subtractor. 1.4 Boolean Algebra and Logic Gates: Basic logic gates and functions, Truth Tables. Synthesis of logic functions with and-or-not gates, NAND gates, NOR gates.	15
2	2.1 Karnaugh Maps: Minimization with Karnaugh Maps up to 2/3/4 variables. 2.2 Combinational Circuits: Multiplexer, De-multiplexer, Encoders, Decoders 2.3 Sequential Circuits: S-R flip- flop, D flip-flop, JK flip- flop, Race-around condition, Master – slave JK flip- flop, T flip-flop. Application of flip- flops. 2.4 Counters: Asynchronous counter, Synchronous counter, Ring counter, Johnson counter. 2.5 Shift Registers: Introduction, parallel and shift registers (SISO, SIPO, PISO, PIPO), Applications of shift registers.	15
3	3.1 Instruction Set: Instruction codes, Computer Instructions, Timing & Control, Instruction Cycles, Memory reference instruction, Input/output & Interrupts. 3.2 Central Processing Unit: General register organization, Stack organization, Instruction format, Data transfer & manipulation, Program control, RISC, CISC. 3.3 Memory organization: Internal organization, Types of memory, RAM, ROM, EPROM technology used for memory construction. Memory organization and operation: Memory expansion; Memory cell; Different types of memory, ROM, PROM.	15

Self-Learning topics (Unit wise)

Sub-Unit	Topic
1.1	Computer Abstractions and Technology: Basic structure and operation of a computer, Functional units and their interaction. Representation of numbers and characters.
2.2	Logic circuits and functions: Combinational circuits and functions: Basic logic gates and functions, truth tables; logic circuits and functions.

Online Resources

Week 1 course content from NOC: Computer Organization and Architecture A Pedagogical Aspect : https://nptel.ac.in/courses/106/103/106103180/ Binary numeration system- https://www.coursera.org/lecture/digital-systems/l0-1-binary-numeration-system-n3mTp What digital systems are? https://www.coursera.org/learn/digital-systems#syllabus
Logic gates- https://www.khanacademy.org/computing/ap-computer-science-principles/computers-101/logic-gates-and-circuits/a/logic-gates

Reference Books: BCS101B Digital System Design

1.	Unit – 1, 2 and 3	1. M. Morris Mano, “Digital Logic and computer Design”, PHI, Sixth Edition. 2. Donald p Leach, Albert Paul Malvino, “Digital principles and Applications”, Tata McGraw, Seventh Edition. 3. R. P. Jain, “Modern Digital Electronics”, Tata McGraw Hill, Edition: 4. Year:2010.ISBN: 9780070669116
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Course Code: BCS101D: Digital System Design

Practical	Digital System Design	Total Credits: 1
Unit 1, 2, 3	Content	No. of Lectures 30 hours
1.	Study and verify the truth table of various logic gates (NOT, AND, OR, NAND, NOR, EX-OR, and EX-NOR).	
2.	Simplify given Boolean expression and realize it.	
3.	Design and verify a half/full adder.	
4.	Design and verify half/full subtractor.	
5.	Design and verify the operation of flip-flops using logic gates.	
6.	Verify the operation of a counter.	
7.	Verify the operation of a 4 bit shift register.	
8.	Design and implement expressions using multiplexers / de-multiplexers.	
9.	Verify the operations of JK Flip-Flop & SR Flip-Flop.	
10.	Design and implement 3-bit binary ripple counter using JK flip flops.	